

Polymorphism

```

Public class Person
{
    Public String name;
    Public int age;
    Public String address;
    Public String nationality;
    Public Person()
    {
    }
}

```

```

Public Person (String name,
                int age, String address,
                String nationality)
{
    this.name = name;
    this.age = age;
    this.address = address;
    this.nationality = nationality;
}

```

Setters and Getters

```

Public class Employee extends Person
{
    Public double Salary;
    Public String Rank;
    Public String Job;
    Public Employee()
    {
    }
}

```

```

Public Employee (String name,
                 int age, String address,
                 String nationality,
                 double salary,
                 String rank,
                 String job)
{
    Super(name, age, address, nationality);
    this.Salary = salary;
    this.Rank = rank;
    this.Job = job;
}

```

```

Public class SalariedEmployee extends Employee
{
    Private double bonus;
    Private double deductions;
    Public SalariedEmployee()
    {
    }
    Public SalariedEmployee (String name,
                             int age, String address,
                             String nationality,
                             double salary,
                             String rank,
                             double bonus,
                             double deduction)
    {
    }
}

```

```

Public SalariedEmployee (String name,
                         int age, String address,
                         String nationality,
                         double salary,
                         String rank,
                         double bonus,
                         double deduction)
{
    Super(name, age, address, nationality);
    this.Salary = salary;
    this.Rank = rank;
    this.Job = job;
    this.bonus = bonus;
    this.deductions = deductions;
}

```

```

Public class test
{
    Public Static void main (String[] args)
    {
        SalariedEmployee S =
        new SalariedEmployee ("Ali", 53,
                              "Alex", "Egyptian", 30000, "Manager",
                              "Engineer", 5000, 1000);
    }
}

```

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name	Ali
age	53
address	Alex
nationality	Egyptian
Salary	30000
Rank	Manager
Job	Engineer
bonus	5000
deductions	1000

Polymorphism

* Overriding
* Final

Edison

Public final String get

```
Employee:
@Override
Public void display()
{
    super.display();
    Sy -- ("name" + name);
    Sy -- ("age" + age);
    Sy -- ("address" + address);
    Sy -- ("nationality" + nationality);
    Sy -- ("Salary" + salary);
    Sy -- ("rank" + rank);
    Sy -- ("Job" + Job);
}
```

```
Public class Person
{
    Public String name;
    Public int age;
    Public String address;
    Public String nationality;
    Public Person()
    {
    }
}
```

```
Public Person (String name,
                int age, String address,
                String nationality)
{
    this.name = name;
    this.age = age;
    this.address = address;
    this.nationality = nationality;
}
```

```
Public void display()
{
    Sy -- ("name=" + name);
    Sy -- ("age=" + age);
    Sy -- ("address=" + address);
    Sy -- ("nationality=" + nationality);
}
```

```
Public class Employee extends Person
{
    Public double Salary;
    Public String rank;
    Public String Job;
    Public Employee()
    {
        Ali
        String name;
        int age;
        String address;
        String nationality;
        double Salary;
        String rank;
        String Job;
    }
}
```

```
Public Employee (String name,
                  int age, String address,
                  String nationality,
                  double Salary,
                  String rank,
                  String Job)
{
    super(name, age, address, nationality);
    this.Salary = Salary;
    this.rank = rank;
    this.Job = Job;
}
```

```
Public void setSalary(double salary)
{
    this.Salary = salary;
}
Public double getSalary()
{
    return Salary;
}
Public String getRank()
{
    return rank;
}
```

```
Public class SalariedEmployee extends Employee
{
    Private double bonus;
    Private double deductions;
    Public SalariedEmployee()
    {
        Ali
        String name;
        int age;
        String address;
        String nationality;
        double Salary;
        String rank;
    }
    Public SalariedEmployee (String name,
                              int age, String address,
                              String nationality,
                              double Salary,
                              String rank,
                              double bonus,
                              double deductions)
    {
        super(name, age, address, nationality, Salary, rank, Job);
        this.bonus = bonus;
        this.deductions = deductions;
    }
}
```

```
Public SalariedEmployee (String name,
                          int age, String address,
                          String nationality,
                          double Salary,
                          String rank,
                          double bonus,
                          double deductions)
{
    super(name, age, address, nationality, Salary, rank, Job);
    this.bonus = bonus;
    this.deductions = deductions;
}
```

```
@Override
Public double getSalary()
{
    return Salary + bonus - deductions;
}
Public String getRank()
{
    return rank;
}
```

```
Public class test
{
    Public static void main (String[] args)
    {
        SalariedEmployee S =
        new SalariedEmployee ("Ali", 53,
                               "Alex", "Egyptian", 30000, "Manager",
                               "Engineer", 5000, 1000);
    }
}
```

```
Sy -- (S.getSalary());
```

```
34000
```

name	Ali
age	53
address	Alex
nationality	Egyptian
Salary	30000
rank	Manager
Job	Engineer
bonus	5000
deductions	1000

Polymorphism

reference type reference variable = new object type () ;
↓
Super class

Subclass

Employee E = new Employee() ✓

Employee E = ~~new Person()~~ X

Person p - new Employee ();
Superclass Subclass

P. Meth
88
surkan

[Handwritten notes:]

name	age	add.	Nation	Salary	Rank	Job
------	-----	------	--------	--------	------	-----

```

Public class Employee extends Person
{
    public double Salary;
    public String Rank;
    public String Job;
    public Employee() { Ali
        String name;
        int age;
        String address;
    }
}

```

Public Employee

double salary;
String rank;
String job;

Super (Ali, 53, 5000, Engineer)

this.salary = salary;
this.rank = rank;
this.job = job;

Public void setSalary (double s)
 {
 this.salary = salary;
 }
 Public double getSalary ()
 {
 return salary;
 }
 Public Pin getRank ()
 {
 return rank;
 }

Public class SalariedEmployee extends Employee
{
 private double bonus;
 private double deductions;
 Public static
 {
 Emp.
 Ali
 String name,
 int age,
 String address,
 String initials,
 new Sa
 Alex

Public Salaried Employee double salary
String rank
Engineer
String Job, double bonus, double deduction
Ali 53 Alex Engineer
(name, age, address, nationality);
Salary, rank, Job
30000 Manager Engineer
this.bonus = bonus;
this.deduction = deduction;
@ override
Public double getSalary()

return Salary + bonus - deduction;

30000 + 5000 - 1000

Public String getrank() {

return rank;

Public class test
ic void main (String[] args)

Employee S =
new SubjectEmployee("Ali", 53,
"Alex", "Egyptian", 30000, "Marriage",
"Engineer", 5000, 1000);
S.set bonus(3000);
sy — (S.get salary());

34000

Sub	name
2	age
3	address
4	nationality
5	salary
6	rank
7	Job
8	bonus
9	deductions
10	